# 🔥 1️⃣ What is Azure Load Balancer?

Azure Load Balancer is:

> Layer 4 (TCP/UDP) load balancing service.

It distributes traffic across multiple VMs.

It works on:

- IP
- Port
- Protocol

❌ It does NOT inspect HTTP headers

❌ It does NOT understand URLs

## 🧠 Types of Azure Load Balancer

| Type | Use Case |
|---|---|
| Public Load Balancer | Internet-facing apps |
| Internal Load Balancer | Private internal apps |

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## 🧠 SKUs (Exam Important)

| SKU | Features |
|---|---|
| Basic | Limited, old |
| Standard | Production-ready |

# 🔥 Core Components of Load Balancer

1️⃣ Frontend IP

2️⃣ Backend Pool

3️⃣ Health Probe

4️⃣ Load Balancing Rule

5️⃣ (Optional) NAT Rule

# 1️⃣ Frontend IP

This is:

👉 The IP clients connect to.

Can be:

- Public IP (for public LB)
- Private IP (for internal LB)

# 2️⃣ Backend Pool

This contains:

👉 VMs or VMSS that receive traffic.

Example:

- VM1 (10.0.1.4)
- VM2 (10.0.1.5)

# 3️⃣ Health Probe (Very Important)

Load balancer checks:

Is VM healthy?

If probe fails → traffic not sent to that VM.

Example:

- TCP 80 probe
- HTTP probe on /health

# 4️⃣ Load Balancing Rule

Defines:

| Field | Meaning |
| --- | --- |
| Frontend Port | Port clients connect to |
| Backend Port | Port VM listens on |
| Protocol | TCP/UDP |

Example:

Frontend 80 → Backend 80

# 5️⃣ NAT Rule

Used for:

👉 RDP/SSH to specific backend VM.

Example:

PublicIP:50001 → VM1:3389

PublicIP:50002 → VM2:3389

# 🔥 LAB – Public Load Balancer

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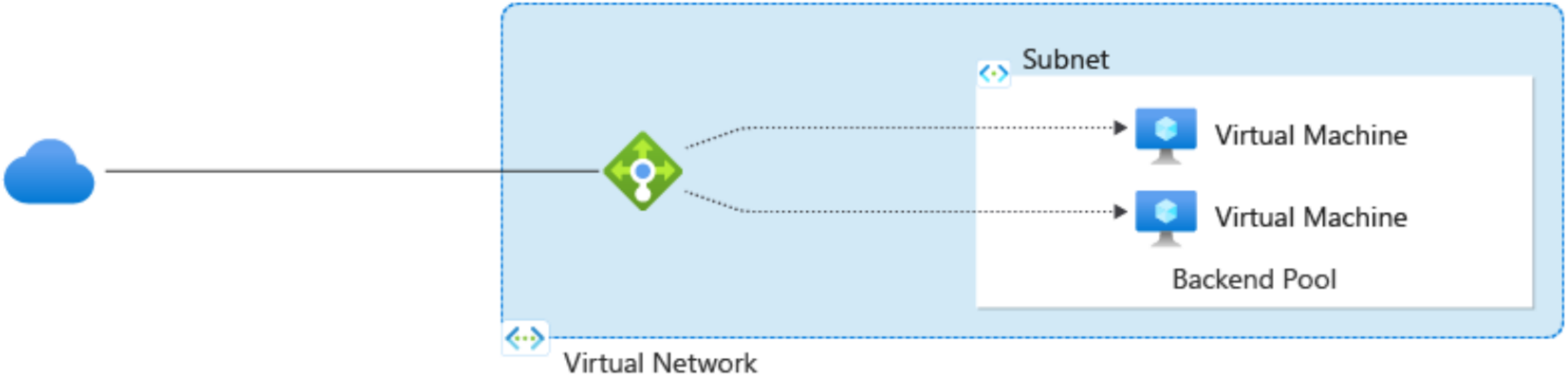
## 🎯 Scenario

2 Web servers behind Load Balancer.

Users access:

```
</> Code

http://PublicIP
```

Traffic distributed across both VMs.


Subnet
Virtual Machine
Virtual Machine
Backend Pool
Virtual Network

## STEP 1 – Create 2 VMs

- Same VNet
- Same Subnet
- Install IIS on both

On VM1:

Create index.html:

Code

```
Hello from VM1
```

On VM2:

Code

```
Hello from VM2
```

| | Name ↑ |
|---|---|
| ☐ | web-1 |
| ☐ | web-2 |

# STEP 2 – Create Load Balancer

Portal → Load Balancer → Create

Choose:

- Type: Public
- SKU: Standard
- Create Public IP

# STEP 3 – Backend Pool

Add:

VM1 NIC

VM2 NIC

# STEP 4 – Create Health Probe

Protocol:

TCP

Port:

80

# STEP 5 – Create Load Balancing Rule

Frontend Port:

80

Backend Port:

80

Associate Health Probe.

Save.

# 🧪 Test

Open browser:

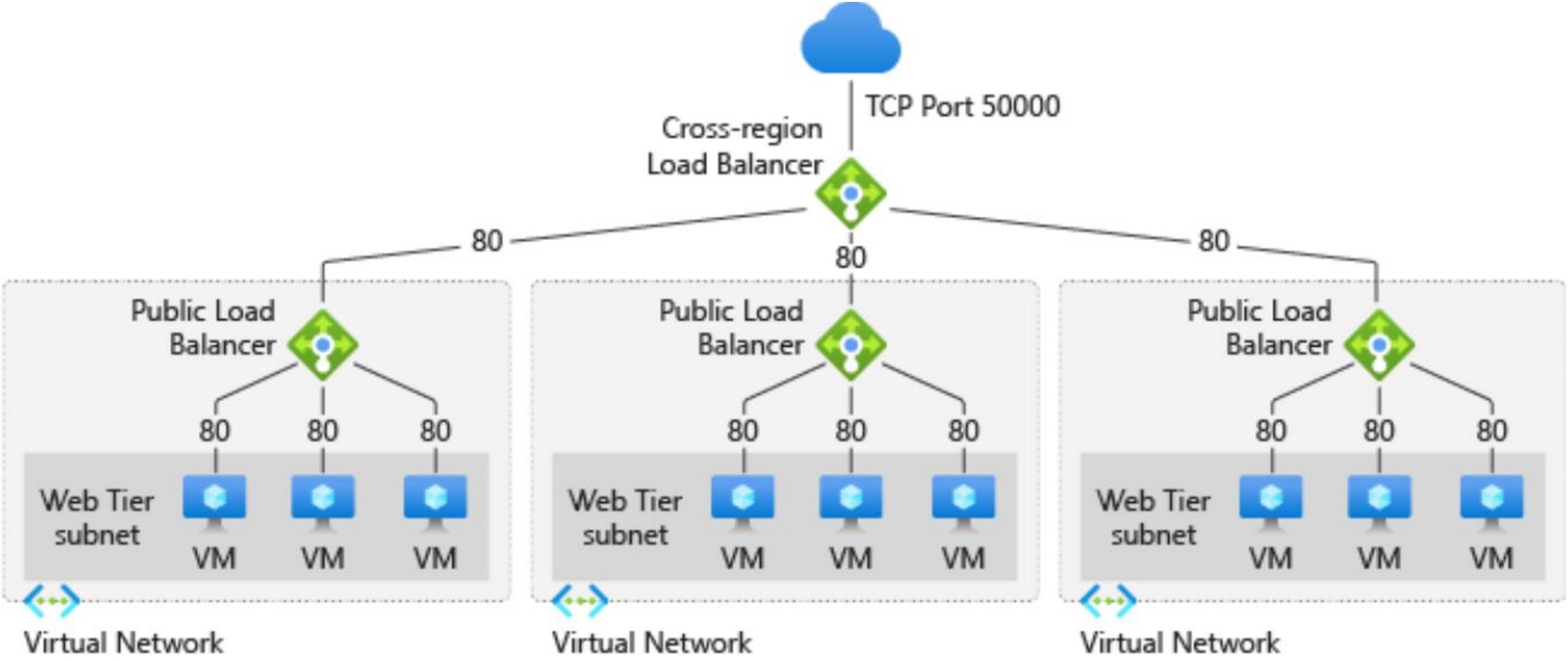
```
</> Code

http://PublicIP
```

Refresh multiple times.

# Global Load Balancer

Azure Standard Load Balancer supports global load balancing enabling geo-redundant high availability scenarios such as:

- Incoming traffic originating from multiple regions.
- Instant global failover to the next optimal regional deployment.
- Load distribution across regions to the closest Azure region with ultra-low latency.
- Ability to scale up/down behind a single endpoint.
- Static anycast global IP address
- Client IP preservation
- Build on existing load balancer solution with no learning curve


TCP Port 50000
Cross-region
Load Balancer
80
80
80
Public Load
Balancer
Public Load
Balancer
Public Load
Balancer
80
80
80
80
80
80
80
80
80
Web Tier
subnet
Web Tier
subnet
Web Tier
subnet
VM
VM
VM
VM
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VM
VM
VM
Virtual Network
Virtual Network
Virtual Network

# Regional redundancy

Configure regional redundancy by seamlessly linking a global load balancer to your existing regional load balancers.

If one region fails, the traffic is routed to the next closest healthy regional load balancer.

The health probe of the global load balancer gathers information about availability of each regional load balancer every 5 seconds. If one regional load balancer drops its availability to 0, global load balancer detects the failure. The regional load balancer is then taken out of rotation.

# Home regions and participating regions

**Home region** is where the global load balancer or Public IP Address of Global tier is deployed. This region doesn't affect how the traffic is routed. If a home region goes down, traffic flow is unaffected.

## Home regions in Azure

- Central US
- East Asia
- East US 2
- North Europe
- Southeast Asia
- UK South
- US Gov Virginia
- West Europe
- West US
- China North 2

# Participating regions in Azure

- Australia East
- Australia Southeast
- Central India
- Central US
- East Asia
- East US
- East US 2
- Japan East

- North Central US
- North Europe
- South Central US
- Southeast Asia
- UK South
- US DoD Central
- US DoD East
- US Gov Arizona
- US Gov Texas
- US Gov Virginia
- West Central US
- West Europe
- West US
- West US 2